

Border mass and the ρ -bound

the prefix mincut as an explicit inequality

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Abstract

The r452 plateau identity $q_* = M_\partial/(M_\partial + 5/7)$ turns the prefix mincut into $\text{err}(\rho) \leq (5/7)/(M_\partial + 5/7)$. This note names both sides. The pack satisfies the elementary identities $q_n^\dagger = S_{n-1}/B_w$ and $\delta_n = (5/7 - \rho_n)/B_w$ (residual $\leq 5 \cdot 10^{-14}$). Hence $q_n < 1$ if and only if $\rho_n < 5/7$, and the first-leak increment is $q_n - q_* = \rho_n/B_w$. M_∂ is the signed folded m_0 (equal to the MAIN defect mass on $k_z \geq 69$) and grows with $\log N$. At every fixed depth $n \in \{20, 40, 80, 160\}$ the ρ -error stays orders of magnitude below the mass margin (RACE_WON). No RH claim. No anti-RH claim.

Status. Ausgang BOUND_PROVED / MASS_CLASSICAL_REDUCED_GROWING / RACE_WON. Theorem 1 and Lemma 1 are the named SATZ. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 The exact pack identities

Write $S_k = \sum_{i \leq k} \rho_i$ and $B_w = S_{n-2} + 5/7$, as in the r362/r369 pack.

Lemma 1 (Schur mass identity). *On every measured depth of $kz5, 17, 116, 136, 137, 230$,*

$$q_n^\dagger = \frac{S_{n-1}}{B_w}, \quad \delta_n = 1 - q_n^\dagger = \frac{5/7 - \rho_n}{B_w},$$

to residual at most $5 \cdot 10^{-14}$ (Gram residual $\leq 10^{-15}$).

These are the r362 telescope $D = B - \sum \rho$ evaluated at prefix depth: $b^\top (I - C)^{-1} b = S_{n-1}$.

Theorem 1 (g -bound). *$q_n^\dagger < 1$ if and only if $\rho_n < 5/7$. Whenever B_w is still the plateau value $B_w^* = M_\partial + 5/7$ (the range $n \leq n_{\text{stab}}$ and the first leak),*

$$q_n^\dagger - q_* = \frac{\rho_n}{B_w} =: g(\rho_n).$$

On the plateau $\rho_n \approx 0$, so $g = 0$ and $q_n^\dagger \equiv q_$.*

kz136 at $n = 176$: $\rho = 0.02540$, $B_w = 5.5337$, $g = 0.00459 = q_{176} - q_*$ (CLIFF).
 kz137 at $n = 176$: $\rho = 3.08 \cdot 10^{-10}$, $g \sim 0$ (CONTINUES). kz17 at $n = 19$: $\rho = 0.1721$,
 $g = 0.05164 = q_{19} - q_*$.

The needed mincut comparison $\text{err}(\rho) \leq (5/7)/B_w^*$ is exactly $\rho_n \leq 5/7$ on a frozen B_w .

2 What M_∂ is

Proposition 1 (Folded m_0). M_∂ is the signed mass of the smooth-border fold. On every $k_z \geq 69$, $M_\partial = \mu_0 - \nu_0 = m_0(\tilde{\mu} - \tilde{\nu})$ of the MAIN fold (equality to 10^{-14}). Source-pure,

$$m_0 = \sum_{j=1}^{L/2} \frac{2}{L} (1 - \cos \theta_j) d_j$$

with the $j = L/2$ half-weight ($\theta_j = 2\pi j/L$, d the window spectral density). This is not a raw ψ -difference of the window edges (those are size $\sim a^2$).

The sister pair has almost the same a (631 vs 641) and $M_\partial = 4.817$ vs 6.436. The source split is the window volume: $N = 1641$ vs 8300, $L = 6562$ vs 33198. $\text{corr}(M_\partial, \log N) = 0.992$. A linear fit $M_\partial = -0.81 + 0.79 \log N$ has rms 0.17; drop-500 is stable. The sample is growing (no ceiling at kz500, $M_\partial = 6.98$). Honest growth law: GROWING, source-identified (CLASSICAL_REDUCED), not a PNT theorem.

3 The race at fixed depth

$\text{err}_n = |\rho_n|/B_w$ and margin = $(5/7)/(M_\partial + 5/7)$.

n	ON max err	ON max ratio	OFF max err	OFF max ratio	verdict
20	$2.2 \cdot 10^{-6}$	$1.6 \cdot 10^{-5}$	$6.8 \cdot 10^{-4}$	$3.2 \cdot 10^{-3}$	WON
40	$3.3 \cdot 10^{-6}$	$2.5 \cdot 10^{-5}$	$1.0 \cdot 10^{-2}$	$5.0 \cdot 10^{-2}$	WON
80	$3.7 \cdot 10^{-6}$	$2.9 \cdot 10^{-5}$	$2.8 \cdot 10^{-3}$	$1.3 \cdot 10^{-2}$	WON
160	$1.9 \cdot 10^{-6}$	$1.5 \cdot 10^{-5}$	$7.6 \cdot 10^{-4}$	$5.8 \cdot 10^{-3}$	WON

ON = $n \leq n_{\text{stab}}$. Once $n_{\text{stab}}(k) > n$ the error collapses to machine-zero while the margin stays in $[0.09, 0.21]$. The r452 moment rate $a^{-1.90}$ is consistent with this collapse (degreewise latch $\Rightarrow \rho_n = 0$). Even OFF-plateau, $\text{err} \ll \text{margin}$ on the measured set. RACE_WON.

4 Scramble

Proposition 2 (Mass definition survives). *Theta-jitter of kz17 (seed 451) moves the MAIN signed mass and kills the plateau (r451). The definition $M_\partial = \int d(\partial\tilde{\mu})$ is unchanged (the smooth border is not the jittered MAIN). The rate picture ($\log N$, $\rho_n \rightarrow 0$) is fold-arithmetic and dies under scramble, as required.*

Claim boundary

Nothing here is evidence for or against the Riemann hypothesis, in either direction. The identities are a finite pack/mass evaluation on measured windows.

References

- [1] Plateau identity; $q_* = M_\partial/(M_\partial + 5/7)$.
- [2] PREFIX_Q_PLATEAU; ρ -exit anatomy.
- [3] Coupled-tau telescope; $D = B - \sum \rho$.
- [4] Prefix mincut; n_{stab} -compression.